

Fermentation Screening Report for Plant-Based Ingredients

Project summary

This internal-style sample report describes a small fermentation screening experiment for plant-based ingredient development. The goal was to compare microbial strains, substrates, and process conditions, then identify variables that could be useful for machine learning models in future screening work.

Screening results

Experiment	Substrate	Strain	pH	Temperature	Duration	Outcome
F-01	wheat bran	Lactobacillus plantarum	6.2	32 C	48 h	protein solubilization reached 18.4 g/L; mild acidic aroma
F-02	soybean meal	Aspergillus oryzae	5.8	30 C	72 h	protein solubilization reached 21.7 g/L; strong enzyme activity
F-03	oat hull hydrolysate	Bacillus subtilis	7.0	37 C	24 h	lower yield of 9.6 g/L; faster pH drift

Useful machine learning variables

For predictive modelling, the most useful features are substrate type, microbial strain, initial pH, fermentation temperature, fermentation duration, moisture content, pretreatment type, inoculum concentration, aeration condition, and measured yield. Important targets include protein solubilization, enzyme activity, final pH, biomass formation, and sensory quality score.

Suggested modelling approach

A first modelling baseline could use random forest regression or gradient boosting to predict yield from the experimental variables. For sequential measurements, a time-series model could use pH and temperature curves. Gaussian process models may also be useful when the number of experiments is small and uncertainty estimates are important for selecting the next experiment.